Appendix D.1 — Answer-Quality Scoring Rubric and Assessment Protocol

Purpose

This document provides detailed scoring criteria and assessment procedures supporting §3.6.3 of the report. It specifies how the five answer-quality measures were applied consistently to the recorded responses from CurriculumAI and ChatGPT.

Metric	Operational definition	Scoring Criteria
Factual accuracy	Technical correctness of substantive explanations and code, assessed independently of corpus support.	0 =Incorrect/ no valid factual answer 1 = Correct core with a material factual error 2 = Substantively correct without identified material errors
Relevance	Extent to which the response directly and sufficiently addresses the question in the intended Scala course context.	0 = Different or irrelevant construct 1=Related but incomplete or insufficiently linked to Scala 2 = Fully addresses essential question components
Citation faithfulness	Whether the actual displayed citations substantiate the factual claims attributed to their cited source passages.	0 = Cited material does not support associated claims 1 = Partially supported 2 = All assessable cited relationships substantively support N/A= No citations
Unsupported-claim rate	Percentage of distinct, assessable factual claims not substantiated anywhere in	(Unsupported factual claims U ÷ Total assessable factual claims in the response N) × 100

	the complete fixed course corpus.	N/A= where N=0 or genuinely unverifiable evidence flagged for review
Source transparency	Clarity with which the response identifies its information origin or declares that it is external to the course corpus.	0 = No identifiable provenance 1= Indicated but incomplete/ambiguous provenance 2= Inspectable source reference OR explicit external/no-course grounding disclosure

Table D.1: Operational Answer-Quality Measures and Scoring Criteria

Additional Assessment Rules

Factual Accuracy and Relevance

Factual accuracy was assessed against the expected technical answer, original system response and appropriate programming evidence, independently of support from the course corpus. Where responses contained code or explanations from another programming language, factual correctness was assessed according to that language.

Relevance was evaluated separately in the intended Scala context. A technically correct explanation of a transferable concept could receive full relevance credit without Scala-specific code. However, explanations involving different programming constructs or unrelated interpretations could receive lower relevance scores. Missing essential answer components primarily affected relevance, whereas substantive technical errors affected factual accuracy.

Citation Faithfulness

Citation faithfulness was assessed by matching the citations actually displayed in each response with their associated factual claims and examining the referenced passages, examples or diagrams. Sources used to prepare reference answers were not treated as citations supplied by either system.

Where multiple citations appeared, their supporting relationships were assessed collectively at response level. Responses without displayed citations received N/A rather than zero, regardless of factual correctness. An explicit Tier-2 external-generation notice could satisfy the source-transparency criterion but did not constitute a citation or establish course-corpus grounding.

Unsupported-Claim Assessment

The unit of analysis was a distinct, assessable factual assertion rather than an entire sentence, citation or individual code line. Responses were examined sentence by sentence or bullet by bullet, with compound statements separated into independently checkable claims. Headings, conversational filler and repeated restatements were excluded. Factual assertions within code examples and claimed outputs were assessed without automatically counting every code line as a separate claim.

Each claim was evaluated against the complete fixed course corpus. Explicitly stated information and examples clearly entailed by the corpus were considered supported. Claims lacking corpus substantiation were classified as unsupported, even where factually correct. Missing citations alone did not establish a lack of corpus support. Neither the system's response tier nor the question's predefined support category determined its unsupported-claim classification.

For each response, the number of unsupported claims (U), total assessable factual claims (N), calculation and resulting percentage were recorded. Unresolvable evidence was flagged for review rather than assigned a speculative classification.

Calculation and Aggregation

The unsupported-claim rate for each response was calculated as:

Unsupported-claim rate (%) = (U/N) × 100

where U represents unsupported factual claims and N represents all assessable factual claims.

Where N = 0, the rate was recorded as N/A.

Two aggregate measures were distinguished:

- **Pooled unsupported-claim rate**: (ΣU/ΣN) × 100, giving equal weight to individual assessable claims.

- **Mean response-level rate:** The arithmetic mean of valid individual response rates, giving equal weight to each response.

These measures were reported separately because differences in the number of claims per answer could affect the aggregate results.

Source Transparency

Source transparency was assessed separately from factual correctness, citation faithfulness and corpus support. Inspectable course references and explicit external-generation disclosures were evaluated using Table D.1. An external-origin notice identified the response's provenance but was not treated as equivalent to a course citation or evidence that the response was grounded in the fixed corpus.

The question register, captured responses, source evidence, individual scores and supporting calculations are documented in Appendix B.